

CASE REPORT

PATHOLOGY/BIOLOGY

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Sudden Unexpected Death Due to Left Subclavian Artery-esophageal Fistula Caused by Fish Bone[‡]

ABSTRACT: A 53-year-old woman was admitted to the hospital due to unexpected dizziness and died the following morning. To investigate the cause of death, a forensic autopsy along with histological examination was performed 3 days after her death. The major findings of the autopsy were that a fish bone had pierced the left subclavian artery after perforating the esophagus with 680 mL of blood in the stomach and bloody and tarry contents were present in the intestines, and the cause of death was confirmed to be subsequent hemorrhagic shock. Unfortunately, none of her family realized that she had eaten a fish 4 days before the tragedy until the fish bone was found. The present case is rare and instructive. The histopathological findings of left subclavian artery-esophageal fistula induced by a fish bone can be used as a reference in forensic practice.

KEYWORDS: forensic science, foreign body ingestion, fish bone, left subclavian artery-esophageal fistula, hemorrhagic shock

Accidental ingestion of foreign bodies is a common situation in daily life; most people choose to swallow these foreign bodies and live uneventfully without medical treatment. In addition, a fish bone is the most common food-associated foreign body, especially in Asia with an estimated annual incidence of 13/100 000 people (1–3). Among all types of foreign bodies, esophageal foreign bodies such as fish bones can unfortunately lead to severe fatal diseases such as left subclavian artery-esophageal fistula and thoracic aortic aneurysms under rare circumstances (4). More than 100 cases of aortoesophageal fistula (AEF) caused by esophageal perforation arise from fish bones, whereas only two cases of left subclavian artery-esophageal fistula induced by fish bones were included among all the reported cases (5,6).

Subclavian artery-esophageal fistula is lethal despite the low morbidity. According to previous literature, the disease usually occurs in patients with aberrant right subclavian arteries with an incidence range from 0.5% to 2.3% worldwide (4,5,7). To the best of our knowledge, only six cases of left subclavian artery-esophageal fistula have been reported, and most of the deceased suffered from foreign body ingestion among all the case reports (5,6,8–11).

In the present case, systematic autopsy and histopathological examination were performed to disclose detailed findings and provide extra date. We observed that a fish bone perforated through the esophagus and pierced the left subclavian artery of the deceased patient, which was determined to be the cause of death.

Case History

A 53-year-old woman admitted to a rural hospital due to unexpected dizzy over the course of 2 days. The medical history review demonstrated that she was a hypertensive patient with intermittent medication therapy. She was pale in appearance during the physical examination, and her blood pressure was 110/76 mmHg. Blood examination revealed severe anemia: red blood cells amounting to $2.99 \times 10^9/L$ ($3.5\text{--}5.0 \times 10^9/L$), hemoglobin level amounting to 9.0 g/dL (11.0–15.0 g/dL), and a 28.40 hematocrit level (37–48%). As it was a rural hospital, no extra auxiliary examinations were performed except the electrocardiogram examination which showed no abnormalities. The primary diagnosis was uncertain, and the patient was administered an intravenous infusion to alleviate the symptoms. On the following morning, she was found to be dead on the floor near the bed with blood on her face. One of the family members claimed medical malpractice and request a medico-legal autopsy to determine the cause of death.

Autopsy Finding

The autopsy was performed four days postdeath. Both of the conjunctivas were found to be pale in appearance. No abnormalities were found on oral examination. There was no sign of trauma or injury except for several injection points.

The fish bone perforated the esophageal wall 7.0 cm away from the epiglottitis and pierced the left subclavian artery 0.4 cm from the beginning of the left subclavian artery, with

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hemorrhage around the fistula. The size of the esophageal rupture was 0.8×0.3 cm, and the size of the rupture of left subclavian artery was 0.5×0.3 cm. The total length of the fish bone was 3.5 cm, and the fish bone was "S"-shaped. The lengths of the fish bone in the esophagus and left subclavian artery were 1.3 and 0.2 cm, respectively (Fig 1b). There was 680 mL of blood in the stomach, and bloody and tarry contents in the intestines. All organs looked anemic in appearance.

Histology of the tissues surrounding the fistula showed infiltration of inflammatory cell dominated by neutrophils. A laminated organization in terms of a laminated thrombus was found in the aorta located around the fistula (Fig 1c). The cause of death was gastrointestinal hemorrhage due to left subclavian artery-esophageal fistula secondary to fish bone ingestion.

Discussion

The clinical symptoms of subclavian artery-esophageal fistula are similar to those of AEF, which was initially described by

Chiari as a triad that includes chest pain, a sentinel hemorrhage, and fatal bleeding after an asymptomatic interval (12). Nonetheless, these clinical symptoms occur in less than half of patients and are more common and specific in AEF caused by foreign bodies than in AEF with other causes (4). The asymptomatic interval between the swallowing of fish bones and bleeding or even death can be hours or even days, which results in profound difficulties in the clinical diagnosis of AEF. In the present case, the clinical presentation of the patient after swallowing the fish bone was atypical and was not taken seriously by the patient, which eventually led to her death.

The esophageal stenosis occurred 7.0 cm away from the esophageal entrance and 25.0 cm away from the incisors at approximately the fourth and fifth thoracic levels, respectively; this level is formed by the intersection of esophagus, aortic arch, and left main trachea, and the compression of these structures makes this area the most vulnerable place of foreign body impaction in the esophagus. Esophageal perforation is also mostly likely to occur at this level (2). Due to the adjacent relationship with the

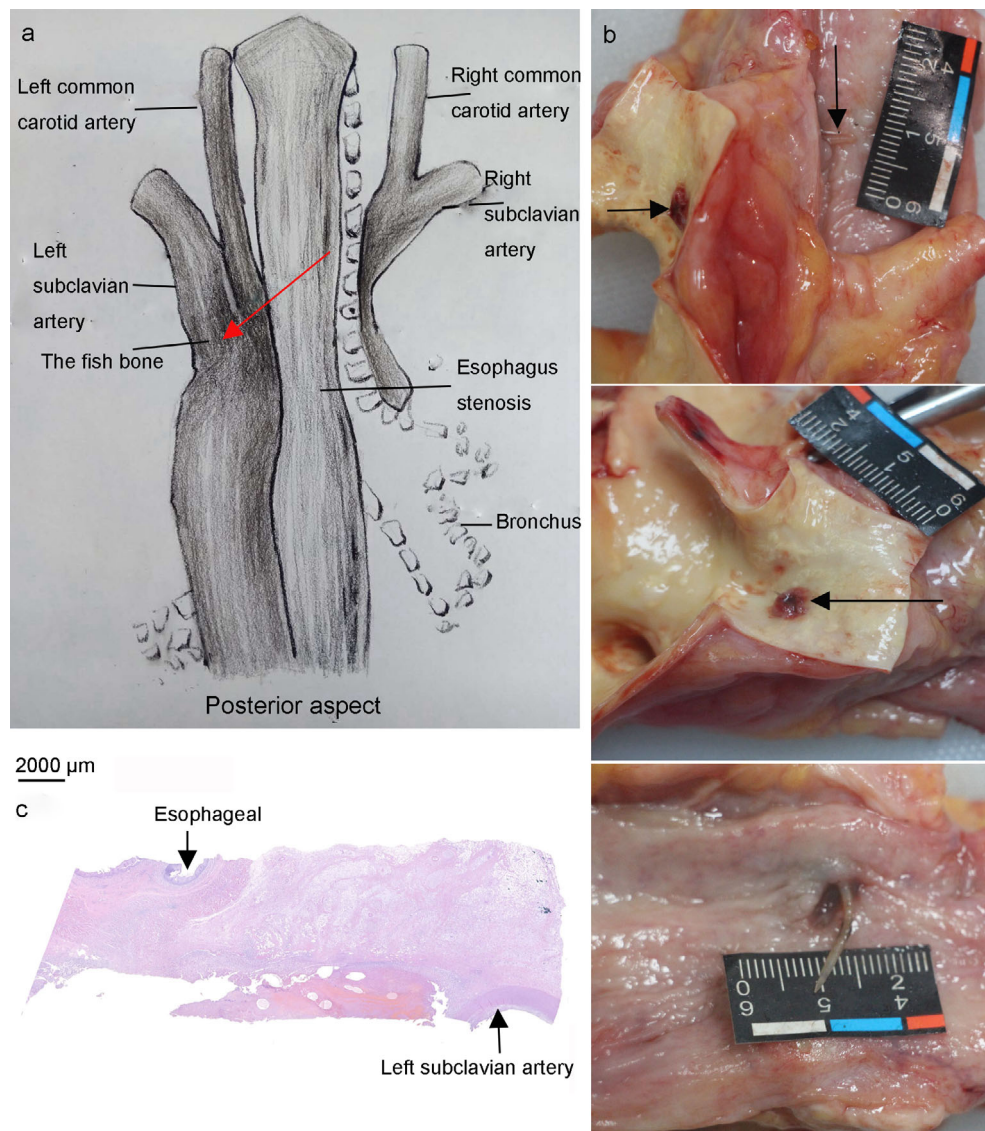


FIG. 1—A fish bone perforated the esophagus and pierced the left subclavian artery: (a) the pattern diagram of this case; (b) arrows show the fish bone and the ruptures in the esophagus and left subclavian artery; (c) histology of the tissues surrounding the fistula showed infiltration of inflammatory cells dominated by neutrophils. Laminated organization in terms of a laminated thrombus was found in the aorta located around the fistula (HE, $\times 6$). [Color figure can be viewed at wileyonlinelibrary.com]

physiological stenosis of the esophagus, the ascending aorta, descending aorta, and the left subclavian artery are the most common sites of vasculo-esophageal fistula (13). Nevertheless, according to the literature, this is the third case of left subclavian artery-esophageal fistula caused by a fish bone.

Perforation of the subclavian artery can be followed by puncture of the fish bone and accompanying infections and inflammatory response. Then, blood clots may form and block the fistula, which prevents hematoma, but the blood clots can become a pseudoaneurysm or mediastinal hematoma. In the present case, only the fistula and surrounding soft tissue were affected, with no pseudoaneurysm or mediastinal hematoma observed. This outcome may be ascribed to the small diameter of the fish bone and the short interval before the patient's death.

Patients who suffer from subclavian artery-esophageal fistula should undergo surgery as soon as possible because patients without surgery have a low change of survival according to previous studies (5). Thus, it was the unadvanced medical condition in a rural area as well as the inexperience of the doctor that led to the tragedy.

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